

REVIEW

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rsc.li/materials-bMonometallic and alloy nanoparticles: a review of
biomedical applicationsElizaveta A. Koroleva, Ilya D. Shabalkin and Pavel V. Krivoschapkin *

Current intrinsic deficiencies in biomedicine promote the rapid development of alternative multitasking approaches. Recently, monometallic and alloy nanoparticles (NPs) have been widely studied for their potential biomedical applications. However, the research mainly focuses on monometallic compounds and metal oxide NPs that have already been studied. In this review, we investigate promising modified mono- and bimetallic NPs for improving the current state of materials science in medicine. It was contended that effective general biomedical applications can be enhanced by intelligent NP design. Particularly, we discuss transition and platinum metal compositions, iron-based and non-iron compounds, along with liquid alloys. Subsequently, we explore the capabilities provided by modifications such as inorganic and organic coatings, polymers, and biomolecules that can invent new NP designs for precise applications, ultimately resulting in an improved patient outcome. We provide a comprehensive assessment of the advantages and limitations of monometallic and alloy nanomaterials and possible solutions to problems that delay their development.

1. Introduction

Because their physicochemical and biological properties are distinctive from those of bulk materials, metallic and alloy nanoparticles (NPs) have been the subject of various studies. These NPs have shown significant potential as functional nanomaterials, and state-of-the-art metallic NPs have found several applications, including in electronics,¹ photonics,² catalysis,^{3–5} drug delivery,^{6–9} and bioimaging,^{10,11} and as magnetic recording devices,¹² biosensors,^{13,14} and nanozymes.^{15,16}

Most metal-based nanomaterials are composed of rare or noble metals, with Au and Ag NPs being the most commonly used and studied. Being noble metals, they are highly biocompatible.^{17–19} Furthermore, Au and Ag NPs also have extraordinary electronic and optical properties, which allow them to be used in biomedical applications such as photothermal therapy (PTT), bioimaging, and biosensing.^{18–23} However, because of insufficient local retention and genotoxic influence, Au and Ag NPs still have biomedical limitations. They stimulate the overproduction of genotoxic radicals, which activates cytokines, kinases, and tumour necrosis factor- α (TNF- α), and leads to inflammation.²⁴ Au NPs tend to have poor thermal stability after long laser irradiation exposure, and recent studies have shown that a six-hour exposure to Ag NPs can lead to a 10-fold increase in the number of reactive oxygen species (ROS) in cells. Moreover, Ag can accumulate in human organs with the prolonged use of Ag-based drugs, causing argyria, a skin condition in which the skin,

eyes, gums, and nails turn a blue-grey colour, particularly those areas exposed to sunlight.²⁵

Since technology and science are dynamic, modern transitional monometallic, liquid metal, and non-iron and iron-based alloy NPs show superior and unique properties. They include catalytic, optical and photoluminescence activities, superparamagnetism, targeting, antibacterial internals. Moreover, all aforementioned properties can be combined into a system based on one nanoparticle for use as a multimodal system with various applications. As such, NPs with new compositions are able to increase the sensitivity of existing diagnosis,^{6,26} drug delivery,^{27,28} oncology treatment,^{29–32} and biosensing methods.^{15,33,34} Generally, these properties are related to the NP interaction with the external stimulus. For instance, superparamagnetic NPs can resonantly respond to the time-varying magnetic fields. NPs may then be clearly defined by MHT. In MHT, on account of Néel and Brown relaxations, the switch of magnetic energy into thermal energy occurs. In the other case, the MRI contrast ability relies on interaction between magnetic NP spins and the surrounding water molecules during this time.³⁵ Meanwhile, there are types of NPs that can be stimulated upon exposure to light. In such plasmonic composites scattered and absorbed visible light excites free electrons in metal atoms to collective oscillation. Next, these NPs may be applied in photoacoustic (PA) imaging and photothermal therapy (PTT).³⁶ Thereby, metallic NPs are unique systems that can clearly exhibit these properties. Since they have a pure composition and the absence of additional compounds, their mechanisms are easier to control and therefore predict behaviour in biological tissues.

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Despite ongoing interest in mono- and bimetallic NPs, their susceptibility to oxidation limits their medical implementation.³⁷ The main application challenge is the high toxicity associated with oxidative stress, which depends on the size of NPs and free radical formation in the body, such as superoxide anions, ROS, reactive nitrogen species (RNS) or hydroxyl radicals.²⁴ Several approaches, including surface shell coating, have been developed to overcome the aforementioned limitations.^{24,37} However, while coatings are able to delay the biotransformation of particles and the release of metal ions, they are prone to degradation. Coating agents not only inhibit free radical generation but also impart new properties to the applied NPs; for instance, Fe₃O₄ coatings confer magnetic susceptibility while SiO₂-coated NPs are photostable and inert. Distinct groups of materials are used to modify the chemical composition of the NP surface, namely natural and synthetic polymers,³⁸ inorganic coatings,³³ ligands, and bioactive molecules.³⁹

A general approach to biomedical applications of mono- and bimetallic NPs is based on their ability to conjugate with different organic ligands. The combination between ligands and size range gives new specificity to cellular uptake as well as simple transport through the blood system. For instance, NPs can effectively penetrate biological barriers, such as the blood–brain barrier (BBB). The superior ability to penetrate barriers is not only due to their size, but also their circulation time in the blood. In such cases, NPs bind to specific transporters and/or receptors expressed on the luminal side of BBB endothelial cells, which increases their circulation time. Additionally, the simplest NP penetration mechanism is based on passive diffusion – the enhanced permeability and retention (EPR) effect.^{40,41} Recently, the synthesis and application of metallic and alloy NPs in biomedicine have not been extensively investigated. The groups of considered NPs include transition and noble metals as monometallic NPs and iron-based, non-iron, and liquid metals as alloy NPs. The main specificity of noble metal NPs in biomedicine is its bioinertness and high chemical stability. Most transition metal and alloy NPs have exclusive magnetic and anti-infective properties.^{30,33,42,43} However, they possess cytotoxic effects in biological environments, to some degree.

This review identifies and discusses promising monometallic and alloy NPs for biomedical innovations. It provides a comprehensive assessment that can accelerate the search for high-performance solutions for biomedical applications and explore their future prospects. The present work highlights the importance of progress made in the characterisation of combined properties of coated NPs for their integration as a one-size-fits-all solution.

2. Monometallic NPs

Monometallic NPs are one of the least studied types of inorganic NPs. Compared to alloy NPs, monometallic NPs have a simple and exact composition. Consequently, it is easier to predict their *in vivo/vitro* behaviour and toxicity and to assess the purity of the final drug formulation. The general classification of the monometallic compounds considered in this review

conditionally includes transition metals and platinum-group elements.

2.1 Transition metal NPs

Cu. Copper is a well-known metal that is important because of its antimicrobial properties and low cost. This metal stimulates the immune system to fight infections, repair injured tissues, and promote healing.⁴⁴ Several studies conducted on copper NPs showed that they have antimicrobial properties similar to those of noble metals.^{42,45} Additionally, copper not only possesses bacteriostatic but also has antifungal properties, inhibiting the growth of harmful microorganisms such as *E. coli*.^{46,47} Its main specificity is based on copper ions that are released from the NP's surface and generate ROS. This destroys bacteria and fungi membranes and induces singlet oxygen-mediated DNA damage and subsequent cell destruction. In this case, Cu NPs demonstrate a 30–45% increase in the cell membrane permeability index of multidrug resistant *S. aureus* (MRSA) compared with untreated controls.⁴² MRSA is an opportunistic pathogen that is resistant to almost all β -lactam antibiotics. Thus, ROS generated on the Cu NP surface prevent biofilm formation and penetrate the bacteria membrane. Cu NPs also exhibit significant apoptosis against U937 and HeLa cells. The MTT assay-based viabilities of U937 and HeLa cells were 15% and 35%, respectively, after treatment with 250 μ M Cu NPs for 24 h.³⁰ However, this mechanism can be detrimental to normal cells. A possible solution to the problem may be controlling NP surface modifications. Thus, Cu NPs could become a new type of antibiotic drug.

Ni. Nickel is very popular in industrial applications. Ni NPs possess high magnetism and bioactivity. Unmodified Ni NPs generate both ROS and RNS⁴⁴ (Fig. 1). According to the literature, Ni NPs exhibit antimicrobial effects against *S. epidermidis*, *S. aureus*, and *E. coli*, and can prevent biofilm formation. To investigate the antimicrobial properties of Ni NPs, Vahedi *et al.*⁴³ added various concentrations of Ni NP suspensions (0.01, 0.05, 0.1, and 1 mg mL^{−1}) to *S. epidermidis* isolates. The results showed that biofilm formation decreased with increasing Ni NP concentration, and the tested concentrations showed low haemolytic effects.^{43,48} Recently, Ni NPs have shown immense promise as a coating for catheter surfaces, which are prone to bacterial colonisation. Fortunately, research into this potential application is ongoing.

Ta. Tantalum is one of the most biocompatible metals.⁴⁹ For a long time, porous bulk Ta has been used to promote the ingrowth of soft tissue, such as blood vessels. Soft tissue formation occurs on the surface of Ta and in its structure.^{49–51} Subsequently, tissue maturation and vascularity increase over a period of time. Because Ta has good biocompatibility (Fig. 2F) and strong optical absorbance, Miao *et al.*⁵¹ developed coated Ta NPs for use as a photoacoustic (PA) contrast agent with high PA intensity. The PA mechanism is based on the excitation of the NPs by laser irradiation followed by high-resolution ultrasound imaging. A simple PA imaging mechanism is shown in Fig. 2A. Ta NPs (107.5 nm) were synthesised using hydrogen arc plasma and then coated with PEG (Fig. 2B–D). The resulting PEGylated



Fig. 1 Schematic illustration of non-modified Cu, Ni, CuNi, and CoNi NPs toxicity mechanisms towards bacteria cells.

NPs are highly stable, have low toxicity, and have PA contrast properties comparable to those of Au NPs. In such applications, Ta can meet various clinical diagnostic requirements due to the different penetration depths of lasers with different wavelengths. Under pulse laser irradiation, Ta NPs significantly improve PA imaging of 4T1 tumours *in vivo* (Fig. 2E). Moreover, the Dox & Ta@CaAlg microsphere system exhibits long-term radiopaque

properties as well as high drug loading capacity and controlled release.²⁷ In this system, Ta NPs act as a contrast agent while Dox is used as a chemotherapeutic drug distributed throughout the CaAlg matrix. The performance of this three-in-one design has been proven *in vivo* and *in vitro* for use in transcatheter arterial chemoembolisation and long-term non-invasive re-examination. Therefore, these metallic NPs can be used in clinical applications,

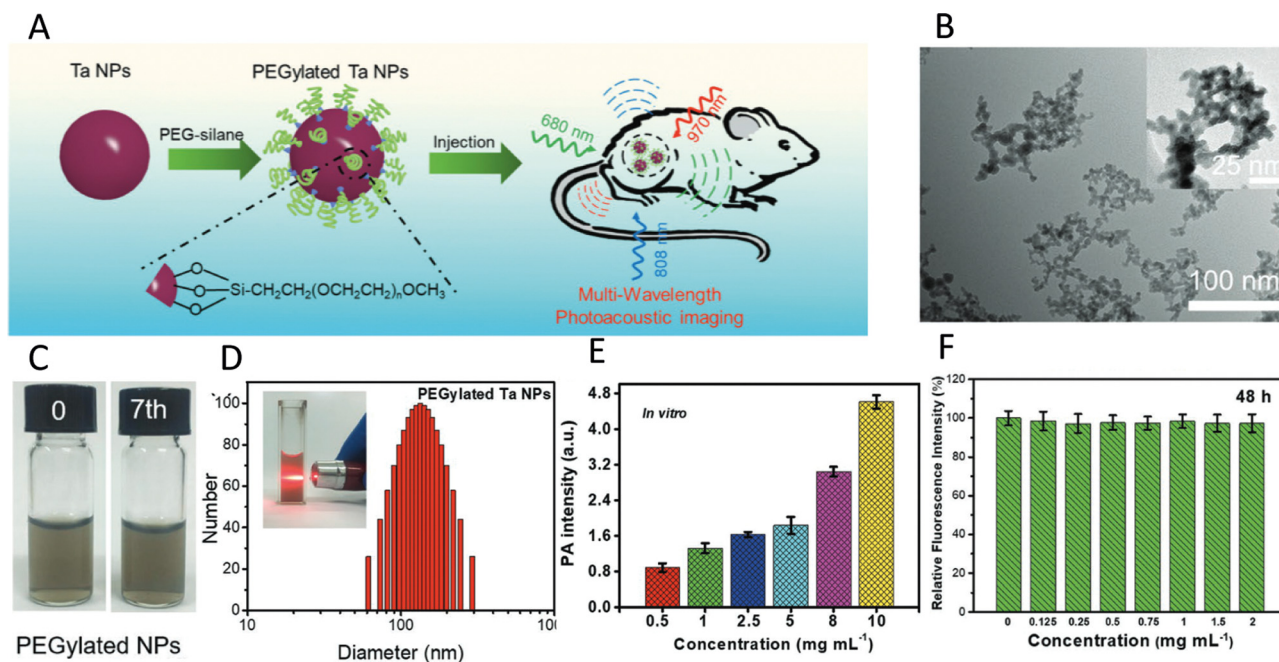


Fig. 2 (A) A scheme of the fabrication process of PEGylated Ta NPs as a metallic photoacoustic contrast agent for multiwavelength photoacoustic imaging of tumors. (B) TEM image of PEGylated Ta NPs. Inset shows the single PEGylated Ta nanoparticle. (C) Digital photographs of PEGylated Ta NPs aqueous dispersions before and after storage for 7 days. (D) DLS size distribution of PEGylated Ta NPs. Inset shows the Tyndall phenomenon of PEGylated Ta NPs dispersion. (E) The corresponding quantitative photoacoustic signal intensity of different concentrations of PEGylated Ta NPs dispersions. (F) Endogenous ROS level in normal HUVECs after incubation with PEGylated Ta NPs for different time durations (48 h). Modified with permission from ref. 51.

especially in imaging, due to their efficacy and biocompatibility. Importantly, Ta-based NPs could become new multimodal systems that do not induce an immune response.

2.2 Platinum NPs

Platinum metals include Pt, Ru, Rh, Os, Pd and Ir; however, so far only Pt, Rh, Pd and Ir have been studied in biomedicine (Fig. 3). Because platinum metals are inert, they are virtually immune to corrosion inside the body. In addition, allergic reactions to these metals are extremely rare. Recent developments showed that platinum metal NPs have the lowest cytotoxic effects. Various platinum-based compounds (*e.g.*, cisplatin, carboplatin) are actively used in cancer chemotherapy. However, commercial drugs have side effects such as vomiting, hypersensitivity reactions, ototoxicity, and nephrotoxicity.⁵² Therefore, more research on the anticancer properties of platinum complexes⁵³ and zero-valent NPs is being conducted.^{38,54,55} These metals also show potential applications in radiation, including X-ray imaging, due to their high electron densities.⁵⁶

Pt. A major advantage of Pt NPs is their biocompatibility – *in vivo* studies have shown no toxicity for 13 weeks.³⁸ Additionally, Pt NPs have inimitable structural, optical, and catalytical properties.^{54,57} Thus, it is worth integrating Pt NPs into biomedicine as nanozymes, nanocarriers, and therapeutic or diagnostic tools. A potential application of these NPs, which have antioxidant properties, was demonstrated by Onizawa *et al.*³⁸ In their paper, they showed that Pt NPs have anti-inflammatory properties and are an effective therapeutic agent for cigarette smoke-related inflammatory lung diseases. Co-precipitation was chosen as the synthesis strategy. K_2PtCl_4 and CTAB solution was heated at 100 °C and then reduced by citrate. Next, Pt NPs were stabilized by polyacrylate (PAA) coating. PAA-Pt NPs can suppress ROS contained and generated in cigarette smoke. They inhibit the apoptosis of A549 cells exposed to cigarette smoke, indicating that oxidative stress has a role in CS-induced cell toxicity. Modified Pt is determined to inhibit A549 cell line death twice as well as *N*-acetylcysteine, which is currently being used. In another example, Zhu *et al.*⁵⁷ proposed the use of porous Pt with high biocompatibility (Fig. 4A) for PTT applications. The authors designed a synergistic anticancer system based on Pt NPs and chemotherapeutic Dox. Porous monometallic NPs were also obtained *via* a reduction method at 70 °C with K_2PtCl_4 precursor

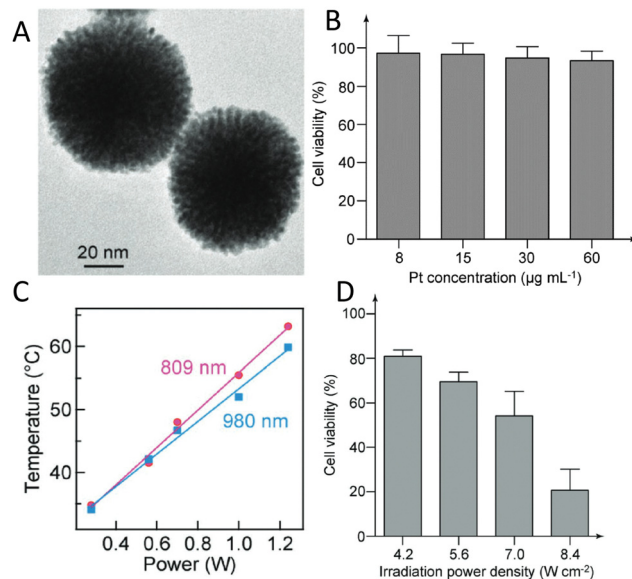


Fig. 4 (A) TEM image of two Pt porous nanospheres. (B) The viability of U-87 MG cells was determined after 24 h exposure to the PDADMAC-coated Pt NPs. The concentration of the Pt nanoparticle sample was varied in the range of 8–60 $\mu\text{g Pt mL}^{-1}$. (C) Photothermal conversion of the Pt NPs in water (2 mL, 240 $\mu\text{g Pt mL}^{-1}$) under 809 and 980 nm laser irradiation. Reached plateau temperatures as functions of the laser power. (D) Cell viability of U-87 MG cells upon PTT. U-87 MG cells were incubated with the Pt NPs at the concentration of 60 $\mu\text{g Pt mL}^{-1}$ for 24 h, followed by the 808 nm NIR laser irradiation at 4.2–8.4 W cm^{-2} for 3 min. The cell viability was determined after 24 h incubation. Modified with permission from ref. 57.

and CTAB stabilizer. Formed during the solubilization $(C_{14}TA)_2\text{-PtBr}_4$ complexes were reduced by ascorbic acid to porous Pt NPs and then were coated layer-by-layer by a PDADMAC-poly(styrene-sulfonate) (PSS) pair. Synthesised Pt NPs exhibit faster and better photothermal activity than Au and iron oxide NPs. Pt NPs allow light absorption in the ultraviolet/visible/near infrared (NIR) regions of the electromagnetic spectrum, which broadens the choice of NIR laser wavelengths (Fig. 4C and D). The aforementioned system demonstrated a modernised combination of plasmonic PTT and chemotherapy modalities for improved cancer treatment. Therefore, highly biocompatible multifunctional Pt NPs can be used to simultaneously treat several distinct medical disorders.

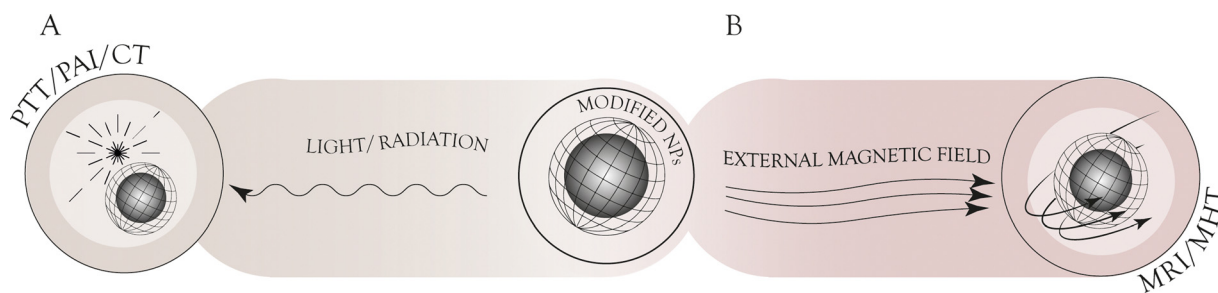


Fig. 3 (A) A schematic illustration of the possible applications (magnetic resonance imaging (MRT) and magnetic hyperthermia therapy (MHT)) of NPs (CuNi, FeCo, CoNi, PtCo, FeNi, FePt NPs) induced by a magnetic field; (B) the schematic illustration of possible applications (PTT, photoacoustic imaging (PAI) and computed tomography (CT)) of NPs (Ta, Pt, Rh, FeCo, FePt NPs) induced by light/radiation.

Rh. Conventionally, rhodium has been used as a catalyst; however, recently, its potential application in biosensing⁵⁸ and PTT for oncological diseases⁵⁵ has garnered significant attention. Importantly, unless NIR-irradiated, Rh-NPs are almost harmless to cell cultures;⁵⁵ hence, many studies have focused on the effectiveness of rhodium in PTT. Rh NPs (5 nm) synthesised *via* polyol reduction showed significant promise as photosensitising agents.⁵⁵ Firstly, RhCl_3 was reduced by ethylene glycol upon heating (190°C) under reflux for 1 h in the presence of polyvinylpyrrolidone (PVP). Next, HeLa cells were exposed to 5 and 10 mg L^{-1} of PVP-Rh NPs and then irradiated with NIR. HeLa cells were exposed to 5 and 10 mg L^{-1} of Rh NPs then irradiated with NIR. As a result, a significant decrease (down to $50 \pm 2\%$) in cell viability was observed. Machuca *et al.*³² showed that NIR irradiation of Rh-exposed cells ultimately died due to the blockage of β -oxidation. This property shows significant anticancer effects during therapy and safety in the post-therapeutic period. The use of Rh NPs as nanozymes has been described by Li *et al.*³⁴ The authors reported that rhodium-based nanozymes can replace some expensive and difficult-to-synthesise enzymes. The enzyme-like activities of Rh nanozymes include peroxidase (POD)-like and catalase (CAT)-like activity. POD-like nanozymes simultaneously decompose hydrogen peroxide (H_2O_2) on the NP surface and generate strong oxidative hydroxyl radicals ($\bullet\text{OH}$). CAT-like nanozyme activity manifests itself in H_2O_2 splitting into O_2 and H_2O . Rh NPs were obtained by the polyol process. $\text{Na}_3\text{RhCl}_6 \cdot 12\text{H}_2\text{O}$ was added to the heated mixture of PVP, KBr and L-ascorbic acid. The reaction solution was set for 3 hours at 140°C . To receive the required nanozyme, Rh-NPs were stirred in DMSO/water ($v/v = 1:4$) with sparfloxacin (SPX) (12 h), then with human serum albumin (HSA) (3 h). This Rh-based nanozyme injection improved the efficacy of the combined rheumatoid arthritis treatment with both ultrasound-driven sonodynamic therapy and alleviated joint hypoxia. After delivery to inflamed joints, the sonosensitiser SPX is released from Rh-nanozymes by

ultrasound irradiation. O_2 radicals produced by Rh NPs suppress the expression of HIF-1 α , thereby facilitating hypoxia and realising anti-angiogenesis in the joints. Additionally, the generated $\bullet\text{OH}$ radicals inhibit synovial hyperplasia and cartilage destruction. Overall, Rh NPs can be used as broad-spectrum anti-cancer drugs as well as inexpensive biosensors for autoimmune disease therapy.

Ir. In medicine, iridium systems have been used as photo agents⁵⁹ and biosensors.⁶⁰ An advantage of Ir NPs is its easy synthesis without surfactants or other solubilising ligands, which usually increase the amount of impurities in the final product.⁵⁶ The biocompatibility of 2–3 nm Ir NPs was confirmed by A. Brown *et al.*⁵⁶ by *in vitro* tests on hepatocyte carcinoma HepG2 and macrophage J774A.1 cell lines. Moreover, the use of iridium, which has a high extinction coefficient and low photobleaching effect, opens up a new approach to biomedicine. Ma *et al.*⁶⁰ reported that 1,5-diaminonaphthalene (DN) and graphene oxide (GO)-modified Ir NPs are a low-cost high sensitivity system that can be used to label C-reactive protein (CRP) antibodies, indicators of *in vivo* inflammation. To obtain a sandwich-type immunosensor Ir NPs must be grown on a GO-DN substrate. This can be achieved by adding $\text{H}_2\text{IrCl}_6 \cdot 6\text{H}_2\text{O}$ to an ethanol-aqueous with GO-DN solution. Next, NaBH_4 solution was added dropwise, and the mixture was uniformly stirred for 4 h. For further antibody NP labeling Ir NPs/GO-DN was dissolved in PBS followed by adding glutaraldehyde. Then, CRP-antibody was added at 37°C for 1 h. Besides, to block non-specific sites bovine serum albumin (BSA) was added. This sandwich-type immunosensor (Fig. 5) consists of three parts: an Au-antibody, CRP antibody, and Ir-antibody. First, the serum was dropped onto the Au-sensor and the antigens bound with their antibodies. Next, the addition of Ir-antibody, which is attached on the top, forms a sandwich structure. Since Ir NPs have CAT- and POD-like activities, H_2O_2 addition causes a redox reaction, allowing the number of antigens to be easily determined by chronoamperometry. This Ir-based electrochemical

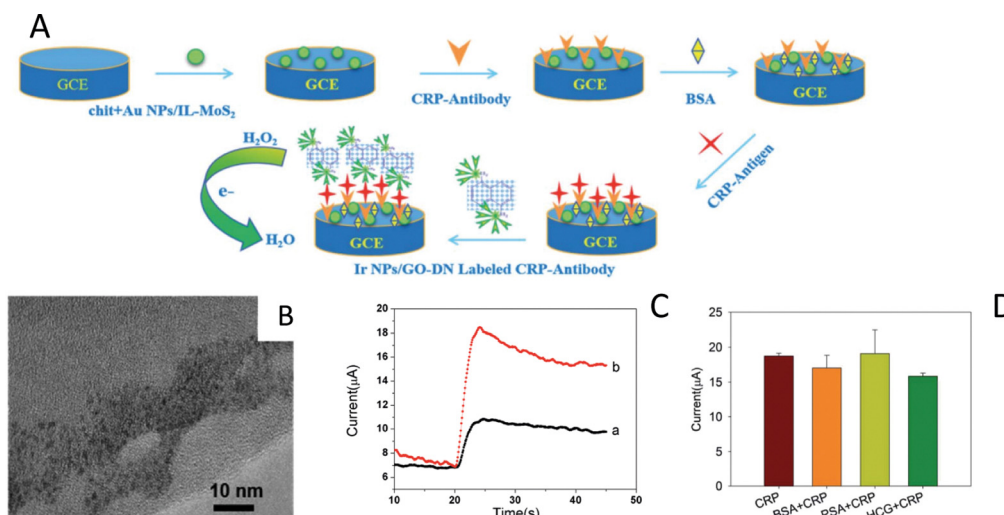


Fig. 5 (A) Flowchart of the preparation of the immunosensor; (B) TEM characterizations of Ir NPs/GO-DN. (C) Comparison of the catalytic effect of (a) GO-DN and (b) Ir NPs/GO-DN to the reduction of 3% H_2O_2 ; (D) the selectivity of the immunosensor. Modified with permission from ref. 60.

immunosensor can be used to analyse real serum samples with significant results.

Wang *et al.*⁶¹ recently developed colorimetric sensor arrays to detect glutathione (GSH) and glucose, by using new Ir-based nanozymes. Ir NPs were obtained by a simple chemical reduction process in 15 minutes. The NP formation reaction proceeds in aqueous phase between $\text{Na}_3\text{IrCl}_6 \cdot x\text{H}_2\text{O}$ and NaBH_4 under 95 °C in the presence of ascorbic acid (Fig. 6A–C). Since Ir NPs demonstrate dual POD- and CAT-like activities, they can catalyse H_2O_2 decomposition. This system shows immense potential as CAT mimics in cancer therapy. Conversely, Ir NPs may consume $\cdot\text{OH}$ radicals, which is atypical of POD mimics, and this POD-activity mechanism is still being researched. However, Ir NPs show great efficiency. Importantly, Ir-based nanozymes monitor blood GSH levels as changes in GSH concentration can indicate diseases such as diabetes, psoriasis, liver damage and Parkinson's disease. By adding GSH to the system, it absorbs the hydrogen peroxide, which leads to an absorbance drop (Fig. 6E). POD activity also allows Ir-based systems to easily detect glucose. As depicted in Fig. 6F, this biosensor detects glucose in 50-fold diluted foetal bovine serum, demonstrating that it can be used in real test systems. Moreover, it has been found that its dual enzymatic activity can be controlled by changes in temperature and pH. As a result, simple inexpensive real Ir-based systems with high sensitivities

and selectivities can be developed. In general, monometallic Ir NPs are still not widely studied, so further developments may reveal new non-biosensing applications.

Overall, monometallic NPs exhibit biological behaviours that will allow the introduction of these systems into medical practice in the near future (Table 1). However, not all compounds have been thoroughly examined due to concerns about their high genotoxicities. On the other hand, noble metal NPs have the lowest toxicity rates because they are inert and oxidation resistant, which reduces adverse effects on cells. More beneficial systems can replace current formulations and help solve existing deficiencies. For instance, although Ir NPs systems exhibit excellent enzymatic and photoactive properties, their practical application is much less developed than Ir-based compounds. Therefore, further evaluation of the efficacy of monometallic NPs and the development of alternative approaches are needed.

3. Bimetallic NPs

In addition to monometallic NPs, alloys with different element ratios are considered promising biomedical materials (Table 2). Their groups can be categorically divided into non-iron- and iron-based compositions. As a rule, bimetallic NPs have incredible magnetic properties, which pique the interest of researchers (Fig. 3). However, their growth mechanism is more complicated than that of pure metal nanocrystals, because controlling the size, shape, composition, and even the degree of atomic ordering of the alloy is much more complex when growing in the solution phase.

3.1 Iron-based bimetallic NPs

FePt. Iron and platinum are two of the most widely used and researched metals in the biomedical field; therefore, the use of their alloys is deeply rooted. Initially, FePt alloys were used in ultra-high-density magnetic recording media and vertical magnetic tunnel junctions due to magnetocrystalline anisotropy, magnetic energy product, and Curie temperature (720 K).^{73,74} It was then discovered that FePt NPs have low toxicity and high X-ray absorption, and therefore can be used as a high biocompatible contrast agent for computed tomography (CT) (Fig. 7).⁷⁵ Furthermore, their superparamagnetic properties make them promising materials in biomedicine,^{76–78} for applications such as T2 magnetic resonance imaging (MRI) contrast agents and in hyperthermic therapy methods.⁷⁹ FePt NPs are usually obtained *via* polyol synthesis. $\text{Pt}(\text{acac})_2$, 1,2-hexadecandiol, and dioctyl ether were mixed and heated to 100 °C under N_2 for 10 min. $\text{Fe}(\text{CO})_5$, oleyl amine and oleic acid were injected at 100 °C and heated to 297 °C for 30 minutes. NP modification with cysteamine was achieved by mixing FePt and cysteamine in ethanol and undergoing further overnight sonication (40–50 °C). A study of the as-synthesised FePt-cysteamine NPs (fcc-FePt-A NPs) in A375M, MCF7 and U2OS cell lines showed their relatively low cytotoxicity and 6-fold better visualisation performance compared to clinical iron oxide NPs.⁷⁹ At a

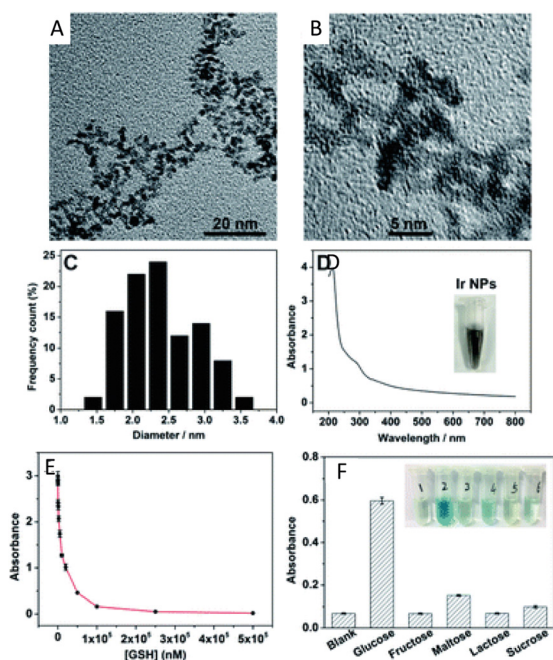


Fig. 6 (A and B) TEM images of Ir NPs at different magnification. (C) Size distribution histogram of Ir NPs. (D) The absorption spectra and digital photos of different colorimetric reaction systems: (a) chromogenic substrate TMB + H_2O_2 , (b) TMB + Ir NPs, and (c) TMB + H_2O_2 + Ir NPs. (E) Dose-response curve for GSH detection at 652 nm. (F) Determination of the selectivity of glucose detection (from left to right: blank, 0.3 mM glucose, 3 mM fructose, 3 mM maltose, 3 mM lactose and 3 mM sucrose). Inset: The color change with the different solutions. Modified with permission from ref. 61.

Table 1 Monometallic NP compositions for biomedical applications

Type	Average size, nm	Coating	Effect	Induction type	Cell line or bacterial stain	Animal model	Application	Ref.
Cu	56	—	ROS	—	<i>E. coli</i> , <i>S. aureus</i>	—	Anti-infective	44
Ni	4–5	—	ROS, RNS	—	<i>S. epidermidis</i> , <i>S. aureus</i> , <i>E. coli</i>	—	Anti-cancer	30
Ta	100	—	ROS, RNS	—	<i>S. epidermidis</i> , <i>S. aureus</i> , <i>E. coli</i>	—	Anti-infective	45
Ta	107, 5	PEG	Ultrasound	NIR	HeLa, HUVEC	4T1 tumour bearing mice	PA imaging	51
Pt	54	Dox & Ta@CaAlg	inhibit A549 apoptosis	—	—	VX2 model	Drug delivery	27
Pt	2	PAA	ROS	—	A549	DBA/2 mice	Antioxidant	38
Rh	72	PDADMAC-PSS	Blockage of β -oxidation	NIR	U-87 MG	—	PTT	57
Rh	5	PVP	CAT- and POD-like activities	NIR irradiation	HeLa	—	PTT	55
Ir	10	SPX/HSA	CAT- and POD-like activities	Ultrasound	—	—	Nanozymes	32
Ir	2–3	CRP antibody	CAT- and POD-like activities	H ₂ O ₂	—	—	Nanozymes	60
	2.5	—						61

concentration of 60 $\mu\text{g mL}^{-1}$, fcc-FePt-A NPs incubated with the aforementioned cell types for 7 days showed $\sim 15\%$ loss in cell viability. However, it is important to note that at concentrations of 30 $\mu\text{g mL}^{-1}$ and below, these NPs have no effect on cell viability, even after incubation periods of up to 7 days. The M_s of 4 nm FePt NPs functionalised with cysteamine is over 100 emu g^{-1} , while the same parameter for currently used iron oxide NPs (6 nm) is 82 emu g^{-1} . This increase affects the specific absorption rate, which shows heating efficiency in weak external alternating magnetic fields, indicating potential application in magnetic hyperthermia treatment. Moreover, the synergy between the radioprotective agent cysteamine and therapeutic FePt can treat cancer while reducing the effect of free radicals on healthy tissues.

Another promising application of FePt is related to photoacoustic (PA) cancer imaging. The mechanism of action is similar to that described for Ta NPs. In their study, Liu *et al.*²⁶ used anti-vascular endothelial growth factor receptor (anti-VEGFR)-conjugated Fe₆₆Pt₃₄ NPs for PA imaging. Anti-

VEGF antibody conjugation allows FePt NPs to target and detect breast tumours in deep tissues. To produce NPs, the aforementioned polyol method was selected. Additionally, the functionalisation procedure was conducted by ligand exchange. Firstly, the anti-VEGFR antibody modified carboxylic group was incubated with ethyl-3-[3-dimethylaminopropyl]carbodiimide hydrochloride (EDC) at 4 °C. Then, *N*-hydroxysuccinimide (NHS) was mixed with this solution. Next, cysteamine-FePt NPs were added into the above mixture and stirred overnight. These FePt-based NPs had an *in vitro* imaging depth of over 5 cm and exhibited targeted delivery and affinity binding of anti-VEGFR ligands. The study also showed that these NPs are easily and rapidly excreted from the tumour site. After 6 h, the FePt NPs were almost entirely metabolised by the liver. The clinical potential of low-toxic FePt NPs is confirmed by its many medical applications, especially in enhanced magnetic hyperthermia treatment and PA theranostics.

FeCo. The synthetic methods and properties of FeCo alloy nanomaterials have been described in several comprehensive

Table 2 Representative bimetallic NPs compositions for biomedical applications

Type	Average size, nm	Coating	Effect	Induction type	Cell line or bacterial stain	Animal model	Application	Ref.
FePt	4	Cysteamine	r_2 relaxivity, free radicals	External magnetic field	A375M MCF7 U2OS	—	MRI, MHT	62
	12, 5	Anti-VEGFR	Ultrasound	NIR irradiation	4T1 breast cancer cells	4T1 breast tumor-bearing mice	PA	26
FeCo	7	C/PL-PEG	r_2 , r_1 relaxivities	External magnetic field	MSCs	Rabbit	MRI	63
	12, 1	C-PEG	Ultrasound	NIR irradiation	4T1 breast cancer cells	4T1 breast tumor-bearing mice	PA	64
	14, 2	—	Free radicals	External magnetic field	—	—	MHT	65
FeNi	64	PEG	Free radicals	External magnetic field	—	—	Magnetic hyperthermia	66
	14	GSH			AGS HepG2 MG63 NCI-H460 SK-MEL-2			67
PtCo	5, 3	—	OXD-like activity	H ₂ O ₂	—	—	Nanozyme	68
	2, 5	PtCo@G@CPB	r_2 relaxivity	pH = 1–3	<i>H. pylori</i>	DBA mice	Antibacterial nanozyme	33
	50	PEG		External magnetic field	4T1, HL 7702	4T1 tumor-bearing mice	MRI	69
CuNi	23	—	Ion release, free radicals	—	<i>E. coli</i> , <i>S. aureus</i>	—	Anti-infective	70
	16	SiO ₂ matrix		Magnetic field	Skin fibroblasts, Caco-2 HeLa	—	Drug delivery	28
	60	PEG			—	—	Magnetic hyperthermia	71
CoNi	60	C	Free radicals	External magnetic field	—	—	Magnetic hyperthermia	72
	300	—			MCF7 C152 HepG2			31

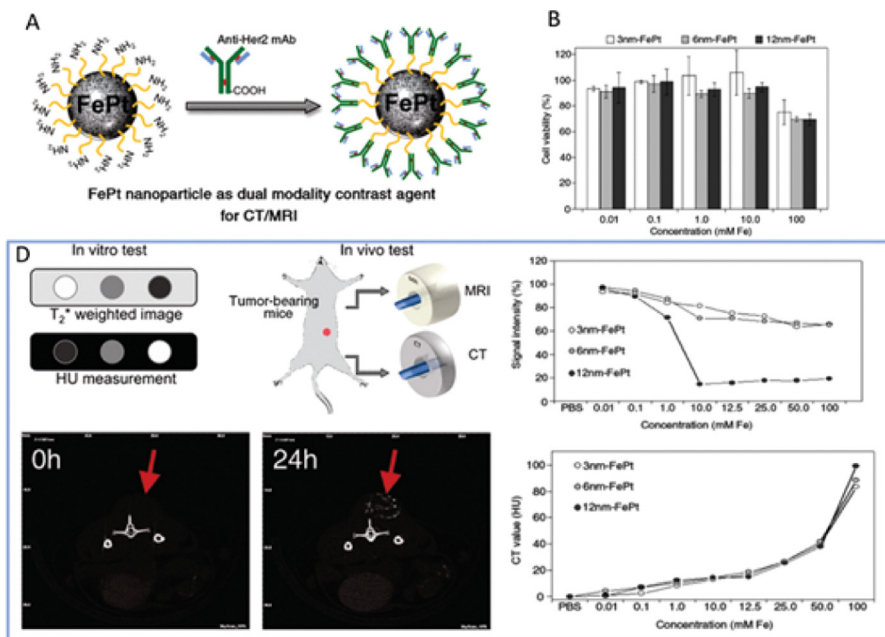


Fig. 7 (A) Schematic illustration showed that cysteamine-capped FePt NPs were prepared and then conjugated with anti-Her2 antibody. The FePt NPs were used as a contrast agent for *in vitro* and *in vivo* MRI and CT evaluation for dual contrast enhancement. (B) The *in vitro* cytotoxicity evaluated by MTT assay in the range of 0.01–100 mM Fe of 3 nm-, 6 nm-, and 12 nm-FePt. (D) The effectiveness of the prepared FePt NPs as CT and MRI contrast agents. Modified with permission from ref. 75.

papers.^{62–65,80–83} Compared to other magnetic NPs, FeCo exhibits excellent magnetic saturation⁶⁴ and low coercivity.⁸² However, these NPs are susceptible to oxidation; therefore, they are often coated to ensure biocompatibility.^{65,82,84} Due to their high magnetic moments, FeCo alloys show great potential for various magnetic applications. To study the contrast MRI effect, Seo *et al.*⁶³ presented a chemical vapour deposition (CVD) method for the synthesis of FeCo with single-shell graphite nanocrystals with PL-PEG functionalisation that are soluble and stable in aqueous solutions. To begin with, $\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ and $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ were loaded onto high-surface-area silica powder by impregnation in a methanol solution of mixed Fe and Co salts. Dried metal-loaded silica was heated to 800 °C under H_2 and then subjected to methane chemical vapour deposition for carbon deposition on the FeCo alloy. Next, to remove the silica substrate, HF was added. The FeCo/GC nanocrystals modification by PL-PEG was performed in the PBS solution (1 h). These NPs are ready for MRI application as contrast agents. Compared to the commercial contrast agents $\gamma\text{-Fe}_2\text{O}_3$ “Feridex” and gadopentetic acid “Magnevist”, 7 nm FeCo NPs showed better r_1 and r_2 relaxivities ($70 \text{ mM}^{-1} \text{ s}^{-1}$ and $644 \text{ mM}^{-1} \text{ s}^{-1}$ at 1.5 T, respectively) for MRI. Therefore, FeCo/graphitic carbon (GC) can reduce the dose of injected MRI contrast agents while demonstrating high contrast efficiency.

In another example, synthesised by a similar CVD method carbon-coated FeCo alloy NPs are used in magnetic- and photo-induced hyperthermia therapy. These NPs possess high near-infrared optical properties, confirmed by significant temperature increases upon NIR irradiation. Carbon coating not only protects FeCo NPs against oxidation but also enhances absorbance in the

NIR-II region; for example, FeCo@C-PEG shows a higher photo-conversion efficiency and stability than Au nanorods. In addition, unlike “Feridex”, which is optically inactive, FeCo@C-PEG NPs can be used in MRI and photoacoustic imaging (PAI).⁶³ *In vivo* experiments showed excellent image-guided thermal ablation of breast cancer under NIR-II irradiation and an alternating external magnetic field.⁶⁴ When treating magnetic hyperthermia, these alloy NPs demonstrate fast heating rates of 30 to 47 °C in 10 min (250 μg). *In vitro/vivo* toxicity assays found that carbon-coated FeCo NPs showed no obvious signs of cytotoxicity.^{63,64} Thus, these results indicate the potential use of these NPs for *in vivo* multi-modal imaging (PAI and MRI) as well as magnetic- and photo-induced hyperthermia therapy.

FeNi. FeNi alloy NPs have received much attention for catalysis.^{3,4} However, recent studies of these bimetallic nanomaterials have shown that they possess high magnetic permeability and anisotropic magnetoresistance, low coercivity, and near zero magnetostriction,⁸⁵ which determine their potential for biomedical applications. FeNi NPs were proposed as magnetic hyperthermia agents with an enhanced heat dissipation capability.⁶⁶ Since the composition, especially the ratio of elements in the nanocrystal, affects its characteristics, the magnetic properties of NPs with different $\text{Fe}_x\text{Ni}_{100-x}$ compositions ($80 \leq x \leq 40$) were analysed.⁶⁶ Alloy NPs with the required elemental ratio were prepared *via* a typical polyol process. Iron(II) chloride and nickel(II) chloride were added to propylene glycol for further dissolution (170 °C). Then, NaOH pellets were added to the solution to act as a reduction agent. Saturation magnetisation (M_s) was found to increase with the amount of nickel in the sample, and the best hyperthermia properties were observed in

the 64 nm $\text{Fe}_{32}\text{Ni}_{68}$ compound ($M_s = 58.24 \text{ emu g}^{-1}$), which showed maximum specific power losses of 241 and 264 W g^{-1} in ethylene glycol and deionised water media, respectively.

In their paper, Mao *et al.*⁶⁷ investigated the biocompatibility of FeNi NPs prepared *via* the same synthesis strategy. Further surface NP modification occurred during the ultrasonic treatment. When modified with various biomolecules (*e.g.*, cysteine (Cys), 5-aminovaleric acid (5AA), and GSH), FeNi NPs were non-toxic *in vitro* toward several cell lines, which was confirmed by toxicity studies for AGS, HepG2, MG63, NCI-H460, and SK-MEL-2 cell lines. The research revealed the low toxicity ($>87\%$ viability) of the modified stable NPs up to a concentration of $50 \mu\text{g mL}^{-1}$.⁶⁷ All modified FeNi NPs exhibit excellent magnetic properties at room temperature; the M_s of 14 nm NPs averaged about 80 emu g^{-1} . This indicates the potential application of highly biocompatible FeNi NPs as a new low-cost method of treating cancer using magnetic hyperthermia.

3.2 Non-iron bimetallic NPs

PtCo. PtCo alloy NPs have high oxidase (OXD)-like activity for selective cancer cell-detection, which can improve the quantitative and rapid colorimetric detection of cancer cells. Furthermore, these superparamagnetic PtCo–aptamer nanoconjugates have affinity for water and show low cell toxicity.⁶⁸ This kind of biosensor does not require H_2O_2 , which catalyses chromogenic agents.⁶⁸ PtCo NPs with an average size of 5.3 nm were obtained by a reduction mechanism (Fig. 8A). In general, the OXD-like activity of these bimetallic NPs is based on the oxidation of substrates with oxygen (Fig. 8D). The PtCo biosensor detects cancer cells with the MUC1 aptamer and catalytically oxidises chromogenic substrates (*e.g.*, TMB, OPT, ABTS). *In vitro* colour change of the chromogenic substrate (Fig. 8B)

occurs at low MCF-7 counts (500 cells). In several experiments, PtCo NPs with a 72:28 ratio were found to show the highest activity, with a maximum absorbance at 652 nm (Fig. 8C). Thus, these nanoconjugates appear to improve OXD-like catalytic activity for cancer cell detection without destructive H_2O_2 .

Zhang *et al.*³³ designed and developed a pH-responsive PtCo based nanozyme for the selective killing of *H. pylori*. This nanozyme consists of a graphene-isolated PtCo nanocrystal functionalised with the bacteria-binding molecule $\text{C}_{18}\text{-PEG}_n$ -Benzeneboronic acid (PtCo@G@CPB). Authors prepared PtCo NPs through a chemical vapor deposition (CVD) method. The method of synthesis is similar to that described for FeCo NPs. PtCo NPs also demonstrate corrosion resistance and show enzyme-like properties only in the acidic environment of the stomach. *In vivo* tests show that under gastrointestinal acid conditions, the OXD-like activity of PtCo@G is activated, catalysing the production of ROS for the selective treatment of *H. pylori* infections. In turn, solution pH affects the dissociative adsorption of O_2 on the PtCo surface and, therefore, its OXD-like activity. Under neutral intestinal conditions, PtCo@G@CPB did not kill any bacteria. However, at pH 7, the nanozyme exhibited minimal toxicity towards commensal bacteria.³³

PtCo NPs can also be used in cancer diagnostics. The incorporation of copper into noble metal NPs imparts important magnetic properties. Yin *et al.*⁶⁹ presented Pt₃Co-PEG NPs (50 nm) as a strong T2 contrast agent in MRI imaging. The synthesis of Pt₃Co was conducted through a polyol method with $\text{Pt}(\text{acac})_2$ and $\text{Co}_2(\text{CO})_8$ as precursors, and oleylamine and oleic acid as growth-controlling surfactants. Surface PEGylation was observed in chloroform for 2 hours. The synthesised NPs possess high r_2 relaxivity (451.2 mM s^{-1}). Even at low concentrations they showed high stability in physiological solutions

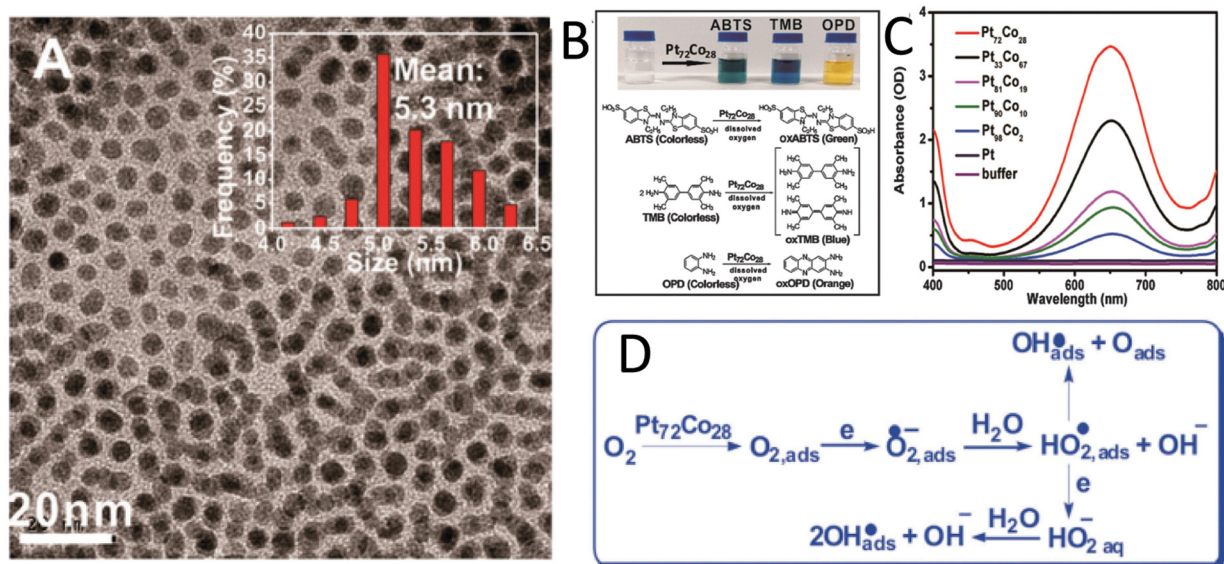


Fig. 8 (A) TEM image of the as-prepared Pt₇₂Co₂₈ NPs (inset: size distribution histogram). (B) The color change and the corresponding reaction mechanisms for the Pt₇₂Co₂₈-catalyzed oxidation of ABTS, TMB and OPD. (C) UV-vis spectra of reaction products during the oxidation of TMB catalysed by Pt–Co NPs with different Pt/Co ratios. (D) Mechanism for the generation of the superoxide anion and the hydroxyl radical. Modified with permission from ref. 68.

and 2.3-fold and 5.4-fold better *in vivo* visualisation than commercial “Resovist” and “Feridex”, respectively. Also, PtCo-PEG NPs (200 mg mL⁻¹) MTT assays showed no significant cytotoxic response toward 4T1, HL 7702, Sk-br-3, and 293T cells. Thus, PtCo NPs promise a solution to existing critical challenges in the clinical treatment of *H. pylori* infections and colorimetric peroxide-free biosensing. In addition, they may prove to be biocompatible T2 contrast agents with excellent *in vivo* visualisation.

CuNi. Most studies on CuNi NPs have focused on involving selective tissue hyperthermia,^{86–88} antibacterial activity,⁷⁰ and drug delivery systems.²⁸ Their chemical stability, corrosion resistance, strong inhibitory activity, and biocompatibility determine their biomedical applications. Parimaladevi *et al.*⁷⁰ showed that CuNi NPs exhibited the highest bactericidal activity against *S. aureus* and *E. coli*. The studied NPs were synthesized through the solvothermal method. CuSO₄ and NaOH were added to hydrazine and ethylene glycol in which hydrazine acts as a reducing agent and ethylene glycol acts as a solvent. The next step included transferring the prepared solution to a Teflon lined stainless steel autoclave with further treatment to 150 °C for 5 hours. The obtained CuNi NPs have a cytotoxic efficiency similar to Cefixime towards both Gram-positive (*S. aureus*) and Gram-negative (*E. coli*) bacteria. ROS that form on the NP surface enhance the inhibitory activity of CuNi. The generated free radicals disrupt the bacterial membrane and cause cytoplasmic leakage, resulting in bacterial death. In addition, positively charged Cu²⁺ and Ni²⁺ change the local pH and conductivity of an environment favourable to bacterial growth. Cu²⁺ ions bind to DNA molecules and disrupt the DNA helical structure by cross-linking. Meanwhile, Ni²⁺ ions displace zinc from the secondary binding site of the metalloprotein fructose-1,6-bisphosphate aldolase (FbaA). Thus, CuNi NPs can be used as highly effective antibiotics.

In another example, Cu_{32.5}Ni_{67.5} NPs in a silica matrix were used as a drug delivery system.²⁸ This system was synthesized *via* the sol–gel method. Sol was prepared by Ni(NO₃)₂·6H₂O, Cu(NO₃)₂·3H₂O and citric acid dissolution in water for 15 minutes. The sol was dried (3 days) at room temperature. Then, gel was calcinated at 500 °C for 24 h in air with further reduction in the H₂/Ar atmosphere for 24 h at 850 °C. The received system can deliver, model fluorescent drugs, for example RHO6G, to human cells. Under an external magnetic field, these NPs can be used to target local tumours and remain at the target site for a certain period. Increased fluorescence was observed near HeLa cancer cells. Furthermore, CuNi-PEG NPs may induce magnetic hyperthermia due to the already proven *M_s* of NPs (45 emu g⁻¹).⁷¹ The use of superparamagnetic CuNi NPs encapsulated in biodegradable PEG polymers allows for great control of local heating effects, resulting in less damage to normal cells. Thus, the NPs can be used as a safe drug delivery system that can release fluorescent agents into target cells and provide local treatment with magnetic hyperthermia.

CoNi. CoNi compounds are characterised by excellent stability, corrosion resistance, and OXD activity.⁸⁹ Additionally, CoNi NPs have electromagnetic, mechanical, and electrocatalytic properties that can be used in electrical, mechanical and medical applications.^{31,89} Currently, radio-absorbing materials (RAMs)

based on a ferromagnetic NiCo/C nanocomposite, which can reduce interference and provide electromagnetic compatibility, are being rapidly developed.⁷² Inspired by this, Yakushko *et al.*⁷² developed 60 nm NiCo/C NPs that combine the advantages of graphite RAMs to provide dielectric and magnetic losses. These characteristics could find new applications in oncological magnetic hyperthermia treatment. Firstly, NiCo NPs were synthesised by the IR pyrolysis of the NiCl₂·6H₂O, CoCl₂·6H₂O and polyacrylonitrile (PAN) precursors solution in dimethylformamide (DMFA). IR annealing was carried out in 2 steps: pre-annealing (removing the solvent) and main annealing in vacuum. The researchers found that increasing the nickel concentration from 10 to 40 wt% resulted in an increase in *M_s* (from 5.94 to 26.7 A m² kg⁻¹) and a decrease in the coercivity *H_c* (from 386 to 287 Oe).

Sargazi *et al.*³¹ studied the cytotoxic effects of 300 nm CoNi NPs. CoNi NPs (6.25 µg mL⁻¹) had favourable cell-killing effects against MCF7 breast cancer and HepG2 liver cancer cells by the magnetic hyperthermia method. These alloy NPs were prepared *via* a polyol process. NaOH was added to the heated up to 175 °C mixture of NiCl₂·6H₂O and Co(CH₃COO)₂·4H₂O in propylene glycol. A histopathological investigation of CoNi (400 mg mL⁻¹ dose) showed hepatocyte necrosis, hepatocyte vacuolisation, and nuclear pyknosis only in cancerous MCF7 and C152 cells.³¹ This selective accumulation may be due to the EPR effect. Of several CoNi NP compositions (Co:Ni = 40:60, 50:50, 60:40), the highest *in vivo* cytotoxic effect and the maximum magnetisation (about 96 emu g⁻¹) were exhibited by Co_{0.6}Ni_{0.4}. Overall, CoNi NPs are promising therapeutic magnetic hyperthermia agents without cytotoxic effects.

From the examples mentioned above, all bimetallic NPs exhibit properties different from monometallic NPs. Non-iron and iron-based alloy NPs combine unique properties used for various applications, especially enzyme-free biosensing, MRI-guided cancer diagnosis, and therapeutic magnetic and photo hyperthermia treatment (Table 2). However, the synergy of the ratios of two different metals gives rise to new and unique magnetic and optical characteristics that can be developed and applied in the near future.

3.3 Liquid alloy NPs

Low-viscosity liquid metals (LMs) such as gallium, gallium–indium eutectic alloys (EGaIn: 75 wt% Ga, 25 wt% In, and gallium–indium–tin alloys (Galinstan: 68.5 wt% Ga, 21.5 wt% In and 10 wt% Sn, have received increasing attention due to their chemical stabilities and non-toxicities in physiological environments.^{39,90–92} These NPs are obtained by simple synthesis, for example, by ultrasonic methods. Current biomedical applications of EGaIn and Galinstan LMs include therapy and imaging.⁹⁰ Reineck *et al.*⁹³ determined that EGaIn NPs exhibit UV plasmonic properties comparable to plasmonic Au NPs. In general, functionalised EGaIn nanocapsules are able to absorb NIR radiation and generate thermal energy and ROS. Exposure to NIR stimuli can transform or destroy the shapes of LM NPs. Furthermore, under laser irradiation, the X-ray intensity of LM NPs increased dramatically. Thus, these systems can be used

for drug encapsulation, cancer treatment, and targeted labelling for X-ray and PA-enhanced imaging. Inspired by this, Lu *et al.*³⁹ developed an LM-based system for enhanced anticancer therapy and MRI imaging (Fig. 9). EGaIn LM NPs were functionalised with ligands, namely thiolated (2-hydroxypropyl)- β -cyclodextrin (designated MUA-CD) and thiolated hyaluronic acid (designated m-HA), which were used as drug-loading matrices for doxorubicin (Dox) and active target compounds. Once in the acidic environment of the tumour, Dox was released from the modified NP surface, and appeared to have a therapeutic effect. *In vivo* tests confirm high effective cancer visualisation and treatment.

A remarkable property of LM NPs is their fictile liquid-metal core. External forces, such as light irradiation or the application of an alternating magnetic field, on LMs operating in physiological environments allow wireless and non-invasive manipulation.⁹⁴ The usefulness of this property was demonstrated in drug delivery systems.⁹⁵ Coating LM NPs with graphene quantum dots (GQDs) provides better photoenergy absorption for phase separation and simultaneous morphological transformation. A change in morphology from nanospheres to hollow nanorods in the liquid metal phase is followed by physical disruption of the endosomal membrane and drug release. Moreover, light irradiation leads to GQD heating, which provides a therapeutic effect. Thus, the designed LM NPs can be used in drug delivery with increased endosomal escape. Another EGaIn LM-based system approach

was carried out by Yu *et al.*⁹² In their study, magnetically controlled LM NPs combined two therapeutic approaches: drug delivery and magnetic hyperthermia. Using an alternating magnetic field (245 kHz), there was a rapid release of heat on the LM NP surface showing rapid heat due to eddy currents. Moreover, the applied H_f factor is two orders of magnitude smaller than that of other magnetic NPs.

Thus, LM NPs allow the development of new types of wireless NPs due to their high biocompatibility, optical activity, magnetic properties, and significant softness and flexibility. The aforementioned LM NPs can release loaded therapeutic agents when exposed to NIR irradiation and alternating magnetic fields. These impressive properties can facilitate the development of new smart non-contact microdevices for cancer treatment.

4. Important challenges and future prospects

Despite their many advantages, most metals also have limitations. Thus, one of the key parameters in the development of mono- and bimetallic NPs for biomedicine is their stabilisation and coatings. Coatings have a wide range of capabilities, such as targeting and minimising NP surface energy to ensure colloidal stability.^{28,82} However, these are not their only advantages. Coating NPs allows various properties that are not intrinsic to NPs to be manifested. For example, coatings can impart optical activity,⁹⁶ magnetism,⁹⁷ or targeting.^{34,98}

The types of coatings used to perform these functions include inorganic and organic coatings, polymers, and biomolecules. As technology has evolved, some systems now include multiple targeting modalities in a single NP (Fig. 10).

4.1 Surface stability and biocompatibility

Silica (SiO_2) is a promising coating because it imparts inert properties and, therefore, strong protection against oxidation. Moreover, it provides a high surface affinity to water, which is very important for invasive drug administration. Generally, silica coating methods are based on the Stober method. Its mechanism involves amine-catalysed hydrolysis and silicon alkoxide condensation.⁹⁹ The Stober method can be used to control the thickness of the silica coating, which is typically from 1 nm to 2 μm .¹⁰⁰ Because of its high photostability, silica is usually used to coat noble metals and quantum dots. SiO_2 prevents changes in surface shape; therefore, the optical properties remain unaltered. Since SiO_2 coatings reduce thermal resistance between NPs and biological environments, it leads to stronger PA signals and longer detection times.¹⁰¹ Moreover, NP chemical stability is due to the fact that this type of coating is a mechanical barrier between the metal surface and the biological environment. Although silica is often used as a coating, research on the SiO_2 -modification of the aforementioned metallic NPs remains relevant. Silica coatings stabilise noble metal and alloy NPs, allowing further modification of other mono- and bimetallic NPs. Moreover, silica structures can

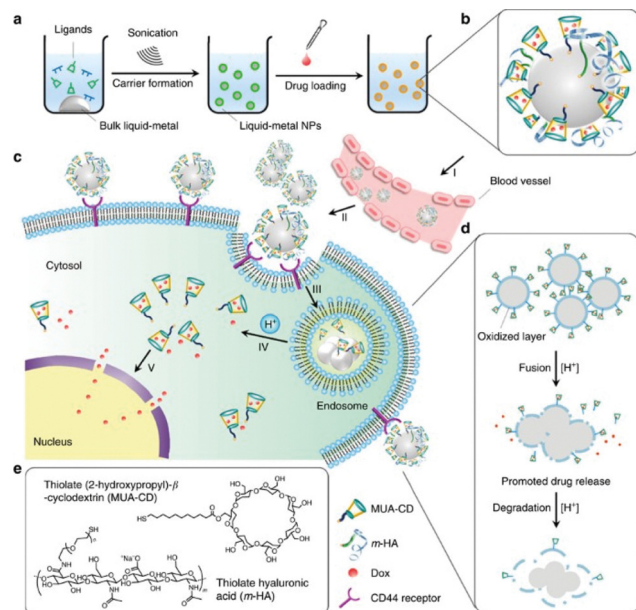


Fig. 9 (a) Preparation route of LM-NP/Dox-L. (b) The main components of LM-NP/Dox-L: thiolated CD with Dox, HA-based targeting motif and an EGaIn core. (c) pH-Responsive delivery of Dox by LM-NP/Dox-L to the nuclei for the targeted cancer therapy. (I) Accumulation of LM-NP/Dox-L at the tumour site through passive and active targeting; (II) specific binding to the overexpressed receptors on the tumour cells; (III) receptor-mediated endocytosis; (IV) acid-triggered fusion of LM-NP/Dox-L and endosomal/lysosomal escape of Dox-containing ligands; (V) accumulation of Dox in the nucleus. (d) Acid-triggered fusion and degradation process of LM-NP/Dox-L. (e) Chemical structures of MUA-CD and m-HA. Modified with permission from ref. 39.

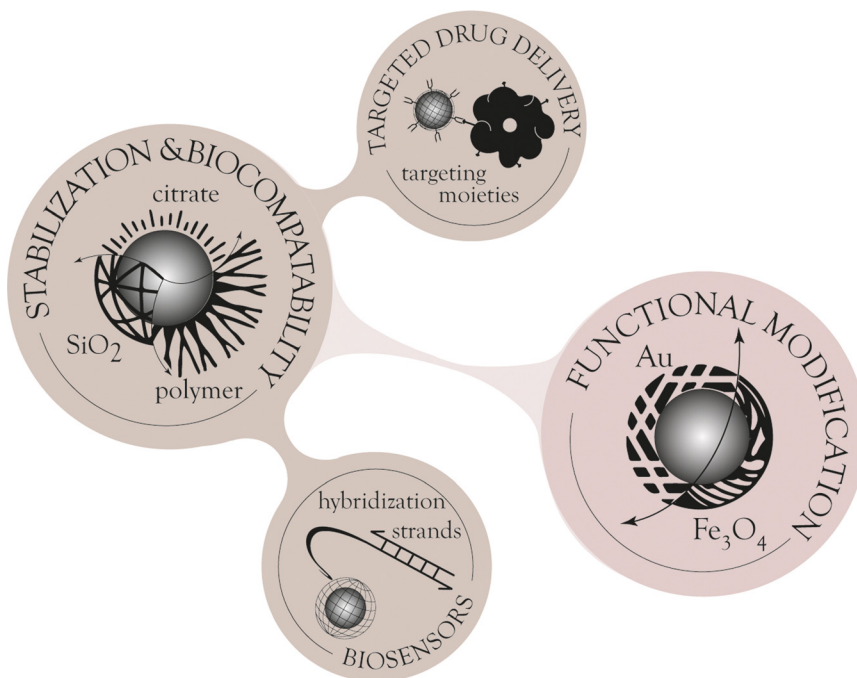


Fig. 10 Overview highlighting some of the surface coatings of mono- and bimetallic NPs.

be classified as dense or mesoporous. The choice of structure is determined by the desired functions, for instance, some drugs can be loaded into mesoporous silica coatings. Compared to mesoporous coatings, dense coatings increase NP chemical stability.¹⁰² Thus, due to silica coatings, unstable and toxic NPs, such as transition metals and their alloys, can find long-term applications without losing their necessary properties.

Polymers are widely used in functionalisation. Steric and electrostatic NP stabilisation occurs *via* the physisorption or chemisorption of polymers on the NP surface.¹⁰³ Moreover, direct encapsulation of hydrophobic NPs by hydrophilic polymers can be performed during synthesis. PEG,^{33,51} PVP,³⁴ and PLGA¹⁰⁴ are the most often used polymers. Many of the aforementioned examples of polymer-modified mono- and bimetallic NPs prove their biocompatibility, stability, and specified properties. The high stability is because of steric stabilisation due to the specific conformation of PEG on the NP surface. High hydrophilicity, chain flexibility, and electrical neutrality leads to PEGylated NPs with high steric stability.^{105,106} In particular, sufficiently active PEG-encapsulated CuNi NPs allow them to be used in magnetic hyperthermia treatment without significant loss of magnetic properties. Polymer modification simultaneously protects the biological environment from the harmful effects of the alloy and metal NPs while protecting the NPs from biodegradation and preserving their enhanced properties.^{33,51,69,71} Biodegradable PEG is considered the “gold standard” for NP coating, as it prevents nanocarriers from interacting with bloodstream components, giving them latent properties and prolonging their plasma circulation time. This, in turn, reduces the dosage and frequency of administration. Similarly, PEGylation prolongs NP plasma and blood half-life by preventing the recognition and eventual excretion of drugs

from plasma by the reticuloendothelial and mononuclear phagocyte systems, and also improves drug penetration into cancer cells *via* the EPR effect.¹⁰⁶ Incidentally, PEG functionalisation generally lowers immunological responses to NPs.¹⁰⁷ Meanwhile, functionalisation with long polymer chains greatly increases the NP size and impairs their tissue permeability, and NPs PEGylated with small-molecular-weight PEG suffer from high protein adsorption (mostly albumin) on the surface.^{108,109} Thus, the length of polymers deposited on the NP surface is determined by their application. In general, polymers, particularly PEG, are promising options for the modification of metal and alloy NPs for the fabrication of long-term biocompatible drugs capable of crossing biological barriers.

In our opinion, citrate shows great potential as a coating because it can also improve the properties of monometallic and alloy NP systems. Citrate is a primary capping agent for NPs. Citrate modification consists of a small citrate acid ionic form with reducing properties or the ability to form complexes with active components. The small size of this type of coating is highly advantageous because it only slightly increases the diameters of the metallic NPs. Unlike polymers and silica, citrate coating is a relatively simple process. Generally, mono- and bimetallic NPs are coated with citric acid during polyol synthesis, which makes ligand adsorption easier. Here, colloidal particles between 2 and 100 nm can be obtained.^{110–112} Citrate coatings stabilise NPs by preventing agglomeration in solution. Citrate modifications are relatively popular; however, unlike stealth polymers, they do not give NPs any additional properties. Fortunately, the citrate surface supports further functionalisation of the colloiddally stable system with biologically active molecules. Hence, this type of coating enables relatively small yet versatile systems to be obtained.

Some surface functionalisations (*i.e.*, silica, PEG, carbon, and citrate coatings) not only confer colloidal stability but also allow further functionalisation for bioconjugation. Thus, NPs with several surface functional groups (*e.g.*, $-\text{NH}_2$, $-\text{COOH}$, $-\text{SH}$) can be easily modified with other ligands.¹⁰² Active ligands deposited on an NP surface can act as effective agents for subsequent biomedical applications (*e.g.*, biosensing, targeted drug delivery, theranostics), resulting in a multifunctional system with good stability and biological activity based on a single NP.

4.2 Targeted drug delivery and biosensing

As mentioned earlier, the simplest distribution mechanism in the biological environment is based on the EPR effect. However, this method may be insufficient for effective therapy or diagnostic methods. This review also found that the EPR effect is not responsible for the large accumulation of NPs in tumours due to vascularisation in human cancer growth. Therefore, NPs modified for targeted delivery are more effective. NP active targeting can be achieved by surface functionalisation with several targeting moieties. These molecules have specificity that allows them to recognise and bind to overexpressed receptors on target cells, triggering cell-penetrating peptide-driven cellular uptake.¹¹³ Examples of such targeting molecules include polypeptides^{26,29} and oligonucleotides.^{15,34,68}

The widespread application of oligonucleotides is due to their high addressability, programmability, and target specificity. Different types of oligonucleotides are used in various fields of research, ranging from biosensing^{34,68} to siRNA delivery.¹¹⁴ Several examples of biosensing tools were previously mentioned. Metallic NPs functionalised with oligonucleotides are simple and fast biosensors with low detection limits. Oligonucleotide (DNA) strands hybridise with specific oligonucleotides in an analyte to produce detectable signals, typically luminescence or fluorescence at specific wavelengths.^{34,60} Metallic NPs act as chemical sensors; they sense complementary DNA strands, which is indicated by a colour change. Furthermore, the covalent attachment of DNA strands to NPs ensures stability. Like peptides and proteins, most NPs cannot be directly functionalised with DNA because they are usually coated. For example, aptamer bio-S2.2 conjugated to streptavidin-coated, PVP-stabilised PtCo NPs allows selective binding of MUC1-positive cancer cells.⁶⁸

Cell penetrating peptides (CPPs) are a class of proteins or peptides that can translocate hydrophilic therapeutic molecules and enhance cellular uptake. CPP-conjugated NPs usually exhibit adsorption-mediated transport. In this mechanism, cationic CPPs interact with negatively charged glycoproteins (proteoglycans) on the cell surface *via* electrostatic interaction, and subsequent endosome formation destabilises the lipid bilayer.⁹⁸ Afterwards, biological microenvironments of diseases have variations in pH, such as the low pH observed in many tumours or the fluctuating pH observed across stages of wound healing.¹¹⁵ Thus, based on metallic NPs pH-sensitive nanoplat-forms have been developed that allow the release of the drug only in specific pH conditions. Moreover, tumour's lymphatic vasculature has abnormalities that form hypoxic environments

(dense ECM and high interstitial pressure) that nanoplat-forms can overcome.^{116,117} Since the NPs were delivered to targeted cells, endogenous triggers (the acidic and hypoxic environment, high enzyme levels of matrix metalloproteinases (MMPs) and other extracellular proteases) can be used to induce its degradation and even drug release. Other exogenous triggers (light, sound waves, radio frequencies and magnetic fields) can also be used for controlled drug release.¹¹⁵ However, peptide or protein coatings have complex deposition and characterisation methods. Although multilayer organic and inorganic coatings significantly increase the particle size.¹⁰³ Yet, CPP coatings are still being studied because appropriate NP encapsulation leads to better stability and optical property retention. For instance, PVP coatings have been used to load human serum albumin (HSA) into Rh nanozymes for rheumatoid arthritis therapy.³⁴

4.3 Functional modifications

Core/shell NPs are currently receiving significant attention due to their multifunctionality. The possibility of combining noble metal and magnetic properties into a single NP allows complementary effects to be realised.^{118,119} For example, both gold and magnetite can be used as shell structures. The gold shell allows metallic NPs to be locally heated by external radiation; however, due to the dissimilar nature of the two surfaces, it is extremely difficult to coat metallic NP coating with Au. Au deposition with iterative hydroxylamine seeding directly on the NP surface results in poor shell morphology. Thus, to obtain monodisperse NPs with dense shell structures, the Au coating can be applied to a bilayer of negatively charged PL-PEG-COOH and positively charged poly-L-histidine (PLH) *via* electrostatic adsorption.⁹⁶ Gold shells also impart surface Raman scattering and tuneable absorption in the NIR region to metallic NPs. Thus, Au coating improves X-ray and PA imaging as well as photothermal and photodynamic therapy.^{120–122} Although many studies have focused on the deposition of gold shells on metal oxide cores, Au functionalisation remains a good option for imparting surface plasmon resonance to optically inactive NPs. In this case, the magnetite core can be replaced by other alloys or metals to obtain the desired optical properties. The combination of a metallic core with magnetic functionality and a plasmonically active gold shell allows these NPs to be used as novel multifunctional biomedical systems. In another example, coating optically active PtCo NPs with gold can lead to a multifunctional system with better imaging capability than commercial Resovist and Feridex due to the magnetic properties of the alloy core. The gold shell not only improves PTT and PAI, but also stability and biocompatibility. Therefore, core-shell NPs composed of various combinations of metals (*e.g.*, Cu, Ni) or alloys (*e.g.*, FePt, FeCo, CuNi, CoNi, and FeNi) coated with Au require further research.

Magnetite coatings have attracted significant interest because they have superparamagnetic properties, and the high M_s of the resulting core-shell structure allows monometallic and alloy NPs to be used in many areas.^{97,123} Passivated magnetite coatings not only stabilise metallic NPs, they also improve metallic NP biocompatibility by reducing toxicity and

corrosivity. In one example, combining a 4 nm FePt core with a 2 nm magnetite shell caused a slight increase in M_s but a strong decrease in H_c . The M_s of pure FePt NPs is approximately 650 emu cm^{-3} ; however, it increases to 1030 emu cm^{-3} when coated in magnetite. Increasing the thickness of the magnetite layer from 0.5 to 2 nm reduced H_c from 13.8 kOe to 60 Oe. These unique characteristics are required for some biomedical applications, such as MRI and MHT.¹²⁴ However, changes in these parameters depend on the initial properties of the core as well as the thickness of the shell. Thus, coating non-magnetic NPs with magnetite can impart magnetic properties to these NPs, expanding their functionality. Meanwhile, when two magnetic materials with significantly different magnetic properties interact in a core-shell structure, strong multiphase structures with unique characteristics are observed.

NP surface chemistry determines NP activity and dispersity. NP toxicity, targeting ability, circulation time in the body, and interaction with proteins and cells depend on the coating. Similarly, different NP/coating combinations allow many distinct multifunctional systems for biomedical applications to be obtained.

5. Conclusion

While many metallic NPs have been used in the development of modern biomedicine, further progress can be made using monometallic and alloy NPs. However, the use of new materials is limited by their toxicity and lack of biocompatibility. Although, due to their unique characteristics and structure, these NPs can replace currently used expensive materials, they are often overlooked. Some modified metallic and alloy NPs show tremendous promise for several specific applications as new multimodal systems. For example, the high selectivities and sensitivities of Ir-, Co-, and PtCo-based nanozymes are ideal for biosensing applications. However, among other mono- and bimetallic NPs, they have doubtful properties for *in vivo*/*in vitro* applications. If we consider the therapy and diagnostics efficiency, Pt and FePt NPs are deeply studied, so they can soon reach real clinical trials. On the other hand, EGaIn and Galinstan liquid alloys require strong attention and deep analysis of experimental data. Likewise, distinct limitations, such as synthesis and stability difficulties, restrict clinical progress of tantalum-based materials with unique tissue regenerating properties.

Most studies on metallic and alloy NPs do not address their biological effects, and their *in vivo* behaviour is not fully understood, making it difficult to realise more biomedical applications for these nanomaterials. Complex oxidation mechanisms and uniform element ratios limit the use of alloy NPs. Similarly, monometallic systems are understudied because they are easily oxidised and have complex formation kinetics. To achieve effective diagnostics and therapy using NPs, it is necessary to learn how to control the adverse effects of materials on organisms. In this way, the unpredictability of NPs in the human body can be reduced with careful composition

control, narrow size distribution, and surface modifications. Importantly, modification ensures long-term stability, target toxicity, and bioaffinity. Coatings can regulate and coordinate changes in the properties of the NP system. Meeting all of these criteria will increase their potential in biomedical applications.

Author contributions

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Conflicts of interest

The authors declare that they have no competing financial interest or personal relationships that could have appeared to influence the work reported in this paper.

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Notes and references

- 1 Y. Li, Y. Wu and B. S. Ong, *J. Am. Chem. Soc.*, 2005, **127**, 3266–3267.
- 2 A. L. Stroyuk, A. I. Kryukov, S. Y. Kuchmii and V. D. Pokhodenko, *Theor. Exp. Chem.*, 2005, **41**, 67–91.
- 3 W. H. Yun, G. Das, B. Kim, B. J. Park, H. H. Yoon and Y. S. Yoon, *Sci. Rep.*, 2021, **11**, 22003.
- 4 S. K. Singh, A. K. Singh, K. Aranishi and Q. Xu, *J. Am. Chem. Soc.*, 2011, **133**, 19638–19641.
- 5 A.-H. Lu and F. Schüth, *Adv. Mater.*, 2006, **18**, 1793–1805.
- 6 K. Guo, Y. Liu, L. Tang and Q. T. H. Shubhra, *Chem. Eng. J.*, 2022, **428**, 131120.
- 7 S.-W. Cao and Y.-J. Zhu, *J. Phys. Chem. C*, 2008, **112**, 12149–12156.
- 8 L. Y. Xia, M. Q. Zhang, C. Yuan and M. Z. Rong, *J. Mater. Chem.*, 2011, **21**, 9020.
- 9 D. Ho, X. Sun and S. Sun, *Acc. Chem. Res.*, 2011, **44**, 875–882.
- 10 N. M. Cordina, N. Sayyadi, L. M. Parker, A. Everest-Dass, L. J. Brown and N. H. Packer, *Sci. Rep.*, 2018, **8**, 4521.
- 11 Y. Du, W. Ren, Y. Li, Q. Zhang, L. Zeng, C. Chi, A. Wu and J. Tian, *J. Mater. Chem. B*, 2015, **3**, 1518–1528.
- 12 W. Chen, Y. Zhou, J. Lu, X. Huang, W. Wu, C. Lin and Q. Wang, *Ceram. Int.*, 2016, **42**, 1114–1121.
- 13 W. Tu, Y. Dong, J. Lei and H. Ju, *Anal. Chem.*, 2010, **82**, 8711–8716.

- 14 M. Hasanzadeh, N. Shadjou and M. de la Guardia, *TrAC*, 2015, **72**, 1–9.
- 15 Y. Chen, P. Wang, H. Hao, J. Hong, H. Li, S. Ji, A. Li, R. Gao, J. Dong, X. Han, M. Liang, D. Wang and Y. Li, *J. Am. Chem. Soc.*, 2021, **143**, 18643–18651.
- 16 C. Cao, H. Zou, N. Yang, H. Li, Y. Cai, X. Song, J. Shao, P. Chen, X. Mou, W. Wang and X. Dong, *Adv. Mater.*, 2021, **33**, 2106996.
- 17 A. C. Powell, G. F. Paciotti and S. K. Libutti, *Cancer nanotechnology, methods in molecular biology*, 2010, pp. 375–384.
- 18 R. Bhattacharya, C. R. Patra, A. Earl, S. Wang, A. Katarya, L. Lu, J. N. Kizhakkedathu, M. J. Yaszemski, P. R. Greipp, D. Mukhopadhyay and P. Mukherjee, *Nanomedicine*, 2007, **3**, 224–238.
- 19 F. Benyettou, R. Rezgui, F. Ravaux, T. Jaber, K. Blumer, M. Jouiad, L. Motte, J.-C. Olsen, C. Platas-Iglesias, M. Magzoub and A. Trabolsi, *J. Mater. Chem. B*, 2015, **3**, 7237–7245.
- 20 O. S. Muddineti, B. Ghosh and S. Biswas, *Int. J. Pharm.*, 2015, **484**, 252–267.
- 21 D. Guo, L. Zhu, Z. Huang, H. Zhou, Y. Ge, W. Ma, J. Wu, X. Zhang, X. Zhou, Y. Zhang, Y. Zhao and N. Gu, *Biomaterials*, 2013, **34**, 7884–7894.
- 22 M. A. Franco-Molina, E. Mendoza-Gamboa, C. A. Sierra-Rivera, R. A. Gómez-Flores, P. Zapata-Benavides, P. Castillo-Tello, J. M. Alcocer-González, D. F. Miranda-Hernández, R. S. Tamez-Guerra and C. Rodríguez-Padilla, *J. Exp. Clin. Cancer Res.*, 2010, **29**, 148.
- 23 J. Wang, J. Li, G. Guo, Q. Wang, J. Tang, Y. Zhao, H. Qin, T. Wahafu, H. Shen, X. Liu and X. Zhang, *Sci. Rep.*, 2016, **6**, 32699.
- 24 A. Ahmet, H. Sipahi and M. Charehsaz, *Nanoparticles toxicity and their routes of exposures*, InTech, Croatia, 2012.
- 25 A. L. S. Chang, V. Khosravi and B. Egbert, *J. Cutan. Pathol.*, 2006, **33**, 809–811.
- 26 Y. Liu, P.-C. Wu, S. Guo, P.-T. Chou, C. Deng, S.-W. Chou, Z. Yuan and T.-M. Liu, *Photoacoustics*, 2020, **19**, 100179.
- 27 J. Zeng, L. Li, H. Zhang, J. Li, L. Liu, G. Zhou, Q. Du, C. Zheng and X. Yang, *Theranostics*, 2018, **8**, 4591–4600.
- 28 J. Stergar, I. Ban, L. Gradišnik and U. Maver, *J. Sol-gel. Sci. Technol.*, 2018, **88**, 57–65.
- 29 K. Perica, A. Tu, A. Richter, J. G. Bieler, M. Edidin and J. P. Schneck, *ACS Nano*, 2014, **8**, 2252–2260.
- 30 G. P. Jose, S. Santra, S. K. Mandal and T. K. Sengupta, *J. Nanobiotechnol.*, 2011, **9**, 9.
- 31 S. Sargazi, M. R. Hajinezhad, A. Rahdar, M. Mukhtar, M. Karamzadeh-Jahromi, M. Almasi-Kashi, S. Alikhanzadeh-Arani, M. Barani and F. Bano, *Appl. Phys. A*, 2021, **127**, 772.
- 32 A. Machuca, E. Garcia-Calvo, D. S. Anunciação and J. L. Luque-Garcia, *Pharmaceutics*, 2021, **13**, 1629.
- 33 L. Zhang, L. Zhang, H. Deng, H. Li, W. Tang, L. Guan, Y. Qiu, M. J. Donovan, Z. Chen and W. Tan, *Nat. Commun.*, 2021, **12**, 2002.
- 34 W. Li, Y. Song, X. Liang, Y. Zhou, M. Xu, Q. Lu, X. Wang and N. Li, *Biomaterials*, 2021, **276**, 121063.
- 35 Q. A. Pankhurst, J. Connolly, S. K. Jones and J. Dobson, *J. Phys. D: Appl. Phys.*, 2003, **36**, R167–R181.
- 36 Z.-Z. J. Lim, J.-E. J. Li, C.-T. Ng, L.-Y. L. Yung and B.-H. Bay, *Acta Pharmacol. Sin.*, 2011, **32**, 983–990.
- 37 J. Lee, G. D. Lilly, R. C. Doty, P. Podsiadlo and N. A. Kotov, *Small*, 2009, 1213–1221.
- 38 S. Onizawa, K. Aoshiba, M. Kajita, Y. Miyamoto and A. Nagai, *Pulm. Pharmacol. Ther.*, 2009, **22**, 340–349.
- 39 Y. Lu, Q. Hu, Y. Lin, D. B. Pacardo, C. Wang, W. Sun, F. S. Ligler, M. D. Dickey and Z. Gu, *Nat. Commun.*, 2015, **6**, 10066.
- 40 J. L. Mikitsh and A.-M. Chacko, *Perspect. Med. Chem.*, 2014, **6**, PMC.S13384.
- 41 M. Shilo, M. Motiei, P. Hana and R. Popovtzer, *Nanoscale*, 2014, **6**, 2146–2152.
- 42 B. R. Shamprasad, R. Lotha, S. Nagarajan and A. Sivasubramanian, *Sci. Rep.*, 2022, **12**, 7061.
- 43 N. Hosseini-Jazani, M. Vahedi, S. Yousefi and M. Ghahremani, *Iran. J. Microbiol.*, 2017, **9**(3), 160.
- 44 R. Magaye, J. Zhao, L. Bowman and M. Ding, *Exp. Ther. Med.*, 2012, **4**, 551–561.
- 45 J. Ramyadevi, K. Jeyasubramanian, A. Marikani, G. Rajakumar and A. A. Rahuman, *Mater. Lett.*, 2012, **71**, 114–116.
- 46 N. Pariona, A. I. Mtz-Enriquez, D. Sánchez-Rangel, G. Carrión, F. Paraguay-Delgado and G. Rosas-Saito, *RSC Adv.*, 2019, **9**, 18835–18843.
- 47 M. Raffi, S. Mehrwan, T. M. Bhatti, J. I. Akhter, A. Hameed, W. Yawar and M. M. ul Hasan, *Ann. Microbiol.*, 2010, **60**, 75–80.
- 48 M. R. Ahghari, V. Soltaninejad and A. Maleki, *Sci. Rep.*, 2020, **10**, 12627.
- 49 C. Balagna, M. G. Faga and S. Spriano, *Mater. Sci. Eng., C*, 2012, **32**, 887–895.
- 50 C.-M. Wu, P.-W. Peng, H.-H. Chou, K.-L. Ou, E. Sugiatno, C.-M. Liu and C.-F. Huang, *J. Alloys Compd.*, 2018, **735**, 2604–2610.
- 51 Z. Miao, P. Liu, Y. Wang, K. Li, D. Huang, H. Yang, Q. Zhao, Z. Zha, L. Zhen and C. Xu, *Small*, 2019, **15**, 1903596.
- 52 H. Renner, G. Schlamp, I. Kleinwächter, E. Drost, H. M. Lüscho, P. Tews, P. Panster, M. Diehl, J. Lang, T. Kreuzer, A. Knödler, K. A. Starz, K. Dermann, J. Rothaut, R. Drieselmann, C. Peter, R. Schiele, J. Coombes, M. Hosford and D. F. Lupton, *Ullmann's Encyclopedia of Industrial Chemistry*, Wiley, 2018, pp. 1–73.
- 53 S. Thangudu, P. Kalluru and R. Vankayala, *Bioengineering*, 2020, **7**, 20.
- 54 P. Puja and P. Kumar, *Spectrochim. Acta, Part A*, 2019, **211**, 94–99.
- 55 A. Machuca, E. Garcia-Calvo, D. S. Anunciação and J. L. Luque-Garcia, *Chem. – Eur. J.*, 2020, **26**, 7685–7691.
- 56 A. L. Brown, H. Winter, A. M. Goforth, G. Sahay and C. Sun, *Nanoscale Res. Lett.*, 2018, **13**, 208.
- 57 X.-M. Zhu, H.-Y. Wan, H. Jia, L. Liu and J. Wang, *Adv. Healthc. Mater.*, 2016, **5**, 3165–3172.
- 58 M. Sadeghi, S. Kashanian, S. M. Naghib, E. Askari, F. Haghiralsadat and D. Tofighi, *Nanotechnol. Rev.*, 2022, **11**, 793–810.

- 59 A. Mesbahi, E. Mansouri and M. Mohammadzadeh, *Pol. J. Med. Phys. Eng.*, 2020, **26**, 225–234.
- 60 Y. Ma, J. Yang, T. Yang, Y. Deng, M. Gu, M. Wang, R. Hu and Y. Yang, *RSC Adv.*, 2020, **10**, 9723–9729.
- 61 Q. Wang, G. Hong, Y. Liu, J. Hao and S. Liu, *RSC Adv.*, 2020, **10**, 25209–25213.
- 62 B. Mehdaoui, J. Carrey, M. Stadler, A. Cornejo, C. Nayral, F. Delpech, B. Chaudret and M. Respaud, *Appl. Phys. Lett.*, 2012, **100**, 052403.
- 63 W. S. Seo, J. H. Lee, X. Sun, Y. Suzuki, D. Mann, Z. Liu, M. Terashima, P. C. Yang, M. V. McConnell, D. G. Nishimura and H. Dai, *Nat. Mater.*, 2006, **5**, 971–976.
- 64 G. Song, M. Kenney, Y.-S. Chen, X. Zheng, Y. Deng, Z. Chen, S. X. Wang, S. S. Gambhir, H. Dai and J. Rao, *Nat. Biomed. Eng.*, 2020, **4**, 325–334.
- 65 L.-M. Lacroix, R. B. Malaki, J. Carrey, S. Lachaize, M. Respaud, G. F. Goya and B. Chaudret, *J. Appl. Phys.*, 2009, **105**, 023911.
- 66 A. Salati, A. Ramazani and M. Almasi Kashi, *Nanotechnology*, 2019, **30**, 025707.
- 67 X. Mao, J. Kwon, E. K. Koh, D. Y. Hwang and J. Lee, *ACS Appl. Mater. Interfaces*, 2015, **7**, 15522–15530.
- 68 S. Cai, C. Qi, Y. Li, Q. Han, R. Yang and C. Wang, *J. Mater. Chem. B*, 2016, **4**, 1869–1877.
- 69 S. Yin, Z. Li, L. Cheng, C. Wang, Y. Liu, Q. Chen, H. Gong, L. Guo, Y. Li and Z. Liu, *Nanoscale*, 2013, **5**, 12464.
- 70 R. Parimaladevi, V. P. Parvathi, S. S. Lakshmi and M. Umadevi, *Mater. Lett.*, 2018, **211**, 82–86.
- 71 J. Chatterjee, M. Bettge, Y. Haik and C. Jen Chen, *J. Magn. Magn. Mater.*, 2005, **293**, 303–309.
- 72 E. v Yakushko, L. v Kozhitov, D. G. Muratov, D. Yu Karpenkov and A. v Popkova, *Russian Microelectronics*, 2020, **49**, 543–553.
- 73 D. Weller and A. Moser, *IEEE Trans. Magn.*, 1999, **35**, 4423–4439.
- 74 Y. F. Xu, M. L. Yan and D. J. Sellmyer, *J. Nanosci. Nanotechnol.*, 2007, **7**, 206–224.
- 75 S. W. Chou, Y. H. Shau, P. C. Wu, Y. S. Yang, D. B. Shieh and C. C. Chen, *J. Am. Chem. Soc.*, 2010, **132**, 13270–13278.
- 76 S.-W. Chou, Y.-H. Shau, P.-C. Wu, Y.-S. Yang, D.-B. Shieh and C.-C. Chen, *J. Am. Chem. Soc.*, 2010, **132**, 13270–13278.
- 77 X. Jiang, S. Weise, M. Hafner, C. Röcker, F. Zhang, W. J. Parak and G. U. Nienhaus, *J. R. Soc., Interface*, 2010, **7**, 5–13.
- 78 T.-L. Tsai, Y.-H. Lai, H. H. W. Chen and W.-C. Su, *Int. J. Nanomed.*, 2021, **16**, 2071–2085.
- 79 S. Chen, L. Wang, S. L. Duce, S. Brown, S. Lee, A. Melzer, S. A. Cuschieri and P. André, *J. Am. Chem. Soc.*, 2010, **132**, 15022–15029.
- 80 J. Alonso, H. Khurshid, V. Sankar, Z. Nemati, M. H. Phan, E. Garayo, J. A. García and H. Srikanth, *J. Appl. Phys.*, 2015, **117**, 17D113.
- 81 Y. Jing, H. Sohn, T. Kline, R. H. Victora and J.-P. Wang, *J. Appl. Phys.*, 2009, **105**, 07B305.
- 82 B. Yang, Y. Wu, X. Li and R. Yu, *Nanomaterials*, 2018, **8**, 154.
- 83 A. Rezazadeh Nochehdehi, N. Revaprasadu, S. Thomas and F. Nemavhola, *Design, Fabrication, and Characterization of Multifunctional Nanomaterials*, Elsevier, 2022, pp. 361–372.
- 84 A. Hütten, D. Sudfeld, I. Ennen, G. Reiss, W. Hachmann, U. Heinzmann, K. Wojczykowski, P. Jutzi, W. Saikaly and G. Thomas, *J. Biotechnol.*, 2004, **112**, 47–63.
- 85 Y. Chen, X. Luo, G.-H. Yue, X. Luo and D.-L. Peng, *Mater. Chem. Phys.*, 2009, **113**, 412–416.
- 86 E. Panizon, J. A. Olmos-Asar, M. Peressi and R. Ferrando, *Phys. Chem. Chem. Phys.*, 2015, **17**, 28068–28075.
- 87 G. Ferk, J. Stergar, M. Drofenik, D. Makovec, A. Hamler, Z. Jagličić and I. Ban, *Mater. Lett.*, 2014, **124**, 39–42.
- 88 S. Araújo-Barbosa and M. A. Morales, *J. Alloys Compd.*, 2019, **787**, 935–943.
- 89 Q. Sheng, D. Liu and J. Zheng, *New J. Chem.*, 2016, **40**, 6658–6665.
- 90 S. A. Chechetka, Y. Yu, X. Zhen, M. Pramanik, K. Pu and E. Miyako, *Nat. Commun.*, 2017, **8**, 15432.
- 91 S.-Y. Tang, D. R. G. Mitchell, Q. Zhao, D. Yuan, G. Yun, Y. Zhang, R. Qiao, Y. Lin, M. D. Dickey and W. Li, *Matter*, 2019, **1**, 192–204.
- 92 Y. Yu and E. Miyako, *iScience*, 2018, **3**, 134–148.
- 93 P. Reineck, Y. Lin, B. C. Gibson, M. D. Dickey, A. D. Greentree and I. S. Maksymov, *Sci. Rep.*, 2019, **9**, 5345.
- 94 S. A. Chechetka, Y. Yu, X. Zhen, M. Pramanik, K. Pu and E. Miyako, *Nat. Commun.*, 2017, **8**, 15432.
- 95 Y. Lu, Y. Lin, Z. Chen, Q. Hu, Y. Liu, S. Yu, W. Gao, M. D. Dickey and Z. Gu, *Nano Lett.*, 2017, **17**, 2138–2145.
- 96 Y. Jin, C. Jia, S.-W. Huang, M. O'Donnell and X. Gao, *Nat. Commun.*, 2010, **1**, 41.
- 97 S. J. A. Figueroa, S. J. Stewart, T. Rueda, A. Hernando and P. de la Presa, *J. Phys. Chem. C*, 2011, **115**, 5500–5508.
- 98 M. Malhotra and S. Prakash, *Curr. Nanosci.*, 2011, **7**, 81–93.
- 99 J.-G. Oh and H. Kim, *Curr. Appl. Phys.*, 2013, **13**, 130–136.
- 100 Z. Li and Y. Zhang, *Angew. Chem.*, 2006, **118**, 7896–7899.
- 101 Y. Chen, C. Xu, Y. Cheng and Q. Cheng, *Photoacoustics*, 2021, **23**, 100284.
- 102 J.-N. Liu, W.-B. Bu and J.-L. Shi, *Acc. Chem. Res.*, 2015, **48**, 1797–1805.
- 103 N. Erathodiyil and J. Y. Ying, *Acc. Chem. Res.*, 2011, **44**, 925–935.
- 104 L.-M. Guo, X.-M. Xu, D. Zhao, X.-G. Cai and B. Zhou, *AMB Express*, 2020, **10**, 160.
- 105 D. Vllasaliu, R. Fowler and S. Stolnik, *Expert. Opin. Drug. Deliv.*, 2014, **11**, 139–154.
- 106 Z. Hussain, S. Khan, M. Imran, M. Sohail, S. W. A. Shah and M. de Matas, *Drug. Deliv. Transl. Res.*, 2019, **9**, 721–734.
- 107 C. D. Walkey, J. B. Olsen, H. Guo, A. Emili and W. C. W. Chan, *J. Am. Chem. Soc.*, 2012, **134**, 2139–2147.
- 108 P. Rivera-Gil, D. Jimenez De Aberasturi, V. Wulf, B. Pelaz, P. del Pino, Y. Zhao, J. M. de La Fuente, I. Ruiz De Larramendi, T. Rojo, X.-J. Liang and W. J. Parak, *Acc. Chem. Res.*, 2013, **46**, 743–749.
- 109 B. Pelaz, P. del Pino, P. Maffre, R. Hartmann, M. Gallego, S. Rivera-Fernández, J. M. de la Fuente, G. U. Nienhaus and W. J. Parak, *ACS Nano*, 2015, **9**, 6996–7008.
- 110 E. Cheraghypour, S. Javadpour and A. R. Mehdizadeh, *J. Biomed. Sci. Eng.*, 2012, **05**, 715–719.
- 111 I. V. Zelepukin, A. v Yaremenko, I. N. Ivanov, M. V. Yuryev, V. R. Cherkasov, S. M. Deyev, P. I. Nikitin and M. P. Nikitin, *ACS Nano*, 2021, **15**, 11341–11357.

- 112 T. Zhu, K. Vasilev, M. Kreiter, S. Mittler and W. Knoll, *Langmuir*, 2003, **19**, 9518–9525.
- 113 R. A. Revia, Z. R. Stephen and M. Zhang, *Acc. Chem. Res.*, 2019, **52**, 1496–1506.
- 114 C. M. Alexander, K. L. Hamner, M. M. Maye and J. C. Dabrowiak, *Bioconjug. Chem.*, 2014, **25**, 1261–1271.
- 115 M. J. Mitchell, M. M. Billingsley, R. M. Haley, M. E. Wechsler, N. A. Peppas and R. Langer, *Nat. Rev. Drug. Discov.*, 2021, **20**, 101–124.
- 116 Y. Zan, Z. Dai, L. Liang, Y. Deng and L. Dong, *Drug Deliv.*, 2019, **26**, 1080–1091.
- 117 H. J. Li, *et al.*, *Proc. Natl. Acad. Sci. U. S. A.*, 2016, **113**, 4164–4169.
- 118 A. E. Sokolov, O. S. Ivanova, A. S. Fedorov, E. A. Kovaleva, M. A. Vysotin, C.-R. Lin and S. G. Ovchinnikov, *Phys. Solid State*, 2021, **63**, 1536–1540.
- 119 P. G. Rudakovskaya, E. K. Beloglazkina, A. G. Majouga, N. L. Klyachko, A. V. Kabanov and N. V. Zyk, *Moscow Univ. Chem. Bull.*, 2015, **70**, 149–156.
- 120 A. F. Moreira, C. F. Rodrigues, C. A. Reis, E. C. Costa and I. J. Correia, *Microporous Mesoporous Mater.*, 2018, **270**, 168–179.
- 121 Y.-N. Wu, L.-X. Yang, X.-Y. Shi, I.-C. Li, J. M. Biazik, K. R. Ratinac, D.-H. Chen, P. Thordarson, D.-B. Shieh and F. Braet, *Biomaterials*, 2011, **32**, 4565–4573.
- 122 A. Tomitaka, S. Ota, K. Nishimoto, H. Arami, Y. Takemura and M. Nair, *Nanoscale*, 2019, **11**, 6489–6496.
- 123 M. E. F. Brollo, R. López-Ruiz, D. Muraca, S. J. A. Figueroa, K. R. Pirota and M. Knobel, *Sci. Rep.*, 2015, **4**, 6839.
- 124 L. C. Varanda, M. Imaizumi, F. J. Santos and M. Jafelicci, *IEEE Trans. Magn.*, 2008, **44**, 4448–4451.